

E0123007

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import random

text = """
Artificial intelligence is transforming the world. Machine learning allows computers to learn from data.
Artificial intelligence and machine learning are closely related fields. Data science helps analyze large amounts of information.
"""

words = text.lower().split()

markov_chain = {}

for i in range(len(words) - 1):
    current_word = words[i]
    next_word = words[i + 1]

    if current_word not in markov_chain:
        markov_chain[current_word] = []

    markov_chain[current_word].append(next_word)

def generate_sentence(length):
    if not words:
        return ""
    word = random.choice(words)
    sentence = [word]

    for _ in range(length - 1):
        if word in markov_chain and markov_chain[word]:
            word = random.choice(markov_chain[word])
            sentence.append(word)
        else:
            break
    return " ".join(sentence).capitalize() + "."

def generate_paragraph(num_sentences=5):
    paragraph = []
    for _ in range(num_sentences):
        sentence_length = random.randint(8, 15)
        paragraph.append(generate_sentence(sentence_length))
    return " ".join(paragraph)

print ( "Generated Sentences: \n" )
for _ in range(3):
    print(generate_sentence(random.randint(8, 15)))

print("\nGenerated Paragraph:\n")
print(generate_paragraph())

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print(generate_paragraph())
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Generated Sentences:

Closely related fields. data science helps analyze large amounts of information..
Is transforming the world. machine learning allows computers.
Analyze large amounts of information..

Generated Paragraph:

Learning allows computers to learn from data. artificial intelligence is transforming the world. machine learning. Are closely related fields. data science helps analyze large